9 Using $Q = mc\Delta\theta$,

Rate of heat removed is $\dfrac{Q}{t} = \dfrac{m}{t} c\Delta\theta$

$$\Rightarrow \frac{m}{t} = \frac{\left(\dfrac{Q}{t}\right)}{c\Delta\theta} = \frac{\left(\dfrac{4.0\times10^{11}}{60\times60}\right)}{4200\times8} = \frac{4.0\times10^{11}}{4200\times8\times60\times60} \text{ kg s}^{-1}$$

Answer: A

10 Using $pV = nRT$